

PRINCE AHMED

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Education

Duke University

Class of 2024

- B.S. in Computer Science and Physics minor.
- GPA: **3.966**/4.0 Magna Cum Laude.
- Karsh International Scholar: 1 of 5 out of the student body to receive a full ride merit based scholarship of \$340,000.
- Coursework: ML, Algorithms, Data Science, Biochemistry, Organic Chemistry, Molecular Biology, Genetics.
- A-levels: **4.5 A* and 1 A** A* in Physics, Chemistry, Biology, Maths, EPQ and A in Further Maths.

Research

Light Sleep EEG Deep Learning Detection

Spring 2024

- Leveraged publicly available annotated datasets to develop and train Deep Learning models achieving **92% accuracy**.
- Utilized Discrete Wavelet Transform for feature extraction in both the frequency and time domains.

Neural Model of Ring Attractor

Spring 2022

- Collaborated in team of 2 to create a Python representation of a network model consisting of over **200 neurons**.
- Demonstrated local connectivity is used in the brain of flies to represent flying direction and presented poster.

Restaurant Selection Application

Spring 2023

- Received grant of \$5000 to develop an algorithm to allow groups with diverse dietary restrictions to choose restaurants.
- Crafted a prototype for user testing resulting in **91% user satisfaction** rate across people at Duke and London.

Evolutionary comparison of intron sequences

Spring 2021

- Utilized PCR amplification, gel electrophoresis, and degenerate primers to analyze extracted gene sequences.
- Successfully cloned the gene into E. coli via ligation into pDrive, followed by ampicillin selection and LacZ screening.
- Used **BLAST** to identify conserved regions indicative of potential enhancer binding sites and presented findings.

Projects

Online Multiplayer Card Game

Spring 2024

- Implemented concurrency management for WebSocket connections by using MongoDB to store states, ensuring security.
- Enabled Google OAuth-based login, room creation/joining via share-code, and live gameplay features.
- Configured CI/CD pipeline to run E2E tests and deploy VueJs UI and ExpressJS server via **Kubernetes**.

ML to predict Airline Delay

Autumn 2022

- Implemented Support Vector Machine Learning model on 5 years' worth of US flight data to predict delays.
- Developed web app that predicts delay probability given a flight number and provides historical data.

Robinhood Clone: BigBucks

Autumn 2023

- Designed **abstractions** to call different APIs to display risk, returns, sharpe ratio and efficient frontier.
- Configured robust **PostgreSQL** database, interactive front-end and Flask MVC back-end.

Global Inequality highlighted by Covid Vaccine Distribution

Spring 2021

- Explored effectiveness of Covid vaccines in light of international and intranational inequalities in their rollout.
- Collaborated in group of 5 (Karsh Scholars) to present findings to Duke merit scholars.

Work Experience

Rumie - Software Engineer Intern

Spring 2024

- Developed backend serving approximately **30,000 users**, to browse, book, and manage dress rentals.
- Created new sorting algorithm to tailor recommendation page to users, increasing marketing **conversion by 40%**.

Finesse Formula LLC - Software Engineer Intern

Summer 2023

- Redesigned home screen and reduced React app load time by **60%** by refactoring several initial NestJS API calls.
- Added multi language support via localization to resolve a common pain point for customers representing 35% in ARR.

Duke Office of Research and Innovation - Data Science Intern

Summer 2022

- Automated review of 20,000+ grant proposal PDFs to find similarities, reducing manual effort from days to 15 minutes.
- Led team of 6 to architect a Transformer Neural Network ([link](#)) pipeline that achieved a **94% accuracy**.
- Managed team and product using Jira, conducted stakeholder interviews, set requirements and acceptance criteria.

NHS Watford General Hospital - Healthcare support worker

2025

- Proactively read medical plans and use a self-made daily to-do list to preempt high-pressure and unpredictability.
- Often being thanked by patients and colleagues for listening and being caring - a patient wrote me a letter saying that I was like a "big brother" during her long-term admission.

Teaching Experience

Discrete Maths for Computer Science

August 2023 – May 2024

- Taught weekly recitation for 30 students - fostered collaboration and lectured more complex material.
- Received incredible feedback from students and was one of the 5 TAs out of 20 that was invited to teach again.

Server Side Web Applications

January 2023 – May 2023

- Held office hours to provide individualized support for Ruby on Rails projects to a class of 70.
- Assisted the delivery of the new course by attending lecture, grading and tailoring projects with the professor.

Skills and Involvements

Volunteering

- Working under Jannatul Ferdous Ivy (BBC top 100 women in 2023) to help with her advocacy for disabilities in Bangladesh - helping with translating and checking English writing.
- 45 hours of medical shadowing with 5 hospitalists by going on rounds and assisting with bedside surgical procedures.
- Duke puppy kindergarten weekly volunteer (2024)
- Part of Duke chapter of APO (service fraternity) weekly volunteering (2024)
- Volunteer weekly at Iglesia Presbiteriana Emanuel church food bank (2023).

Campus Involvements

- Duke Men Rowing Team (2021-2023)
- Langdorm - international student living learning community
- Winner of Senior Physics Challenge - spent 1 week at Cambridge studying Quantum Mechanics

Skills

- Programming Languages: Python, SQL, JavaScript, Swift, Java, R, C/C++, Ruby, TypeScript.
- Technologies: Scikit-learn, PyTorch, Pandas, Linux, React Native, Vue, React, Ruby on Rails, Docker, AWS, Git, Nginx.
- Lab: PCR, gel electrophoresis, degenerate primer design, gene cloning, Microscopy, TLC, ligation and selection.